

ACADEMIC TEXTBOOK SERIES **PROBABILITY**

THEORY A Comprehensive Textbook From Fundamental

Concepts to Advanced Applications in Statistics, Data Science, and Machine
Learning _____ Theory,

*Derivations, Worked Examples,
Applications, Exercises, and Complete Solutions*

Jean Baptiste Habanabakize Data Analytics Curriculum August 13,

2026

“Probability provides a mathematical language for reasoning under uncertainty.”

Preface

Probability theory is the mathematical framework used to describe, quantify, and analyse uncertainty. It provides the foundation for statistical inference, stochastic modelling, risk analysis, machine learning, artificial intelligence, finance, engineering, medicine, and many other areas of science and technology.

This textbook is designed to develop probability from first principles and progressively lead the reader toward more advanced concepts. Particular attention is given to mathematical reasoning, intuitive interpretation, worked examples, and practical applications.

Throughout the book, probability is connected to realistic problems from education, medicine, quality control, finance, reliability engineering, and data science. A unified example based on the *Data Hills Academy* student cohort is used in several chapters to demonstrate how probability concepts can be interpreted using realistic data.

The book follows a progressive learning structure:

1. Fundamental concepts and terminology
2. Sample spaces and events
3. Counting techniques
4. Probability axioms
5. Relationships between events
6. Fundamental probability rules
7. Conditional probability
8. The law of total probability
9. Bayes' theorem
10. Random variables
11. Probability distributions
12. Expectation and variance
13. Limit theorems
14. Applications to statistics and machine learning

Each major topic contains definitions, explanations, mathematical results, examples, applications, exercises, and solutions.

The objective is not simply to memorise formulas, but to understand **why** the formulas work, **when** they should be used, and **how** they can be applied to real-world problems.

Jean Baptiste Habanabakize

Contents

Preface	1
I FOUNDATIONS OF PROBABILITY	5
1 The Landscape of Probability	6
1.1 What is Probability?	6
1.1.1 Interpretations of Probability	6
1.2 The Sample Space and Events	6
1.3 Introducing Our Unified Example: Data Hills Academy	7
1.4 Real-World Application: Quality Control	10
1.5 Exercises	11
2 Counting Techniques	12
2.1 Introduction	12
2.2 The Fundamental Counting Principle	12
2.3 Factorials	12
2.4 Permutations	12
2.5 Combinations	13
2.6 Sampling with and without Replacement	13
2.7 Exercises	13
3 Axiomatic Probability	15
3.1 Probability Spaces	15
3.2 Kolmogorov’s Axioms	15
4 Relationships Between Events	17
4.1 Mutually Exclusive Events	17
4.2 Independent Events	17
4.3 Pairwise vs. Mutual Independence	17
4.4 Collectively Exhaustive Events	18
4.5 Complement of an Event	18

4.6	Real-World Application: Credit Risk	18
4.7	Exercises	18
5	Fundamental Probability Rules	20
5.1	The Complement Rule	20
5.2	The Addition Rule	20
5.3	The Multiplication Rule	21
5.4	Real-World Application: System Reliability	21
5.5	Exercises	22
6	Visualising Probability	23
6.1	Venn Diagrams	23
6.2	Tree Diagrams	23
7	Conditional Probability	25
7.1	Definition and Intuition	25
7.2	The Contingency Table (Probability Form)	25
7.3	Real-World Application: Medical Testing	25
7.4	Exercises	26
8	The Law of Total Probability	27
8.1	Partitions	27
8.2	Statement and Derivation	27
8.3	Real-World Application: Spam Filtering	28
8.4	Exercises	28
9	Bayes' Theorem	29
9.1	The Theorem	29
9.2	Terminology	29
9.3	Real-World Application: Medical Diagnosis	29
9.4	Exercises	30
II	RANDOM VARIABLES AND DISTRIBUTIONS	31
10	Random Variables	32
10.1	Definition and Motivation	32

10.2	Discrete Random Variables	32
10.3	Continuous Random Variables	33
10.4	Exercises	33
11	Discrete Probability Distributions	34
11.1	The Bernoulli Distribution	34
11.2	The Binomial Distribution	34
11.3	The Poisson Distribution	34
11.4	Exercises	35
12	Continuous Probability Distributions	36
12.1	The Uniform Distribution	36
12.2	The Exponential Distribution	36
12.3	The Normal Distribution	36
13	Expectation and Variance	37
13.1	Expected Value	37
13.2	Variance	37
14	Limit Theorems	38
14.1	Law of Large Numbers (LLN)	38
14.2	Central Limit Theorem (CLT)	38
A	Formula Summary	39
B	Glossary of Terms	40
C	Solutions to All Exercises	41

Part I

FOUNDATIONS OF PROBABILITY

Chapter 1

The Landscape of Probability

1.1 What is Probability?

Probability is a numerical measure of the likelihood that an event will occur. It is expressed as a number between 0 and 1, where 0 indicates impossibility and 1 indicates certainty. The study of probability provides the mathematical foundation for dealing with uncertainty, randomness, and risk.

Definition 1.1 (Classical Probability). Let Ω be a finite sample space in which all outcomes are equally likely. For an event $A \subseteq \Omega$, the probability of A is

$$P(A) = \frac{|A|}{|\Omega|}.$$

Worked Example:

Coin Toss: Suppose we toss a fair coin. The sample space is $\Omega = \{H, T\}$. The event $A = \{\text{heads}\}$ has size $|A| = 1$, and $|\Omega| = 2$. Therefore:

$$P(A) = \frac{|A|}{|\Omega|} = \frac{1}{2} = 0.5.$$

So there is a 50% chance of getting heads.

1.1.1 Interpretations of Probability

1. **Classical (theoretical):** Based on symmetry and equally likely outcomes (e.g., coin tosses, dice).
2. **Frequentist (empirical):** Based on long-run frequencies (e.g., historical data on product defects).
3. **Subjective:** Based on personal belief or expert judgment (e.g., probability of a political candidate winning).

In this book, we will primarily work with the frequentist interpretation, since it is most relevant in data analytics and real-world applications.

1.2 The Sample Space and Events

Every random experiment has a set of possible outcomes.

Definition 1.2 (Sample Space). The sample space Ω is the set of all possible outcomes of a random experiment.

Definition 1.3 (Event). An event A is a subset of Ω . We say that A occurs if the outcome of the experiment belongs to A .

Worked Example:

Rolling a Die:

- $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Event $A = \{\text{even number}\} = \{2, 4, 6\}$.
- $|A| = 3, |\Omega| = 6$.
- $P(A) = \frac{3}{6} = 0.5$.

So the probability of rolling an even number is 50%.

1.3 Introducing Our Unified Example: Data Hills Academy

Throughout this book, we will use a single, realistic dataset to illustrate every concept. This approach ensures continuity and helps you see how different probability rules connect.

Before we dive into the numbers, we must define two essential operations on events: **AND** (intersection) and **OR** (union).

Definition 1.4 (Intersection (AND)). The intersection of two events A and B , denoted $A \cap B$, is the event that **both** A and B occur simultaneously.

$P(A \cap B)$ is read as “the probability of A AND B ”.

Definition 1.5 (Union (OR)). The union of two events A and B , denoted $A \cup B$, is the event that **at least one** of A or B occurs. This includes the possibility that **both** A and B occur.

$P(A \cup B)$ is read as “the probability of A OR B ”.

We now present our dataset.

Data Hills Academy – Student Cohort of 1000

- 600 boys (60%), 400 girls (40%).
- Overall: 800 passed (80%), 200 failed (20%).
- Gender breakdown:
 - Boys: 450 passed, 150 failed.
 - Girls: 350 passed, 50 failed.

To make the relationships clear, we organise the data into a **contingency table**.

We define the experiment as selecting one student uniformly at random from the 1,000 students in the cohort. Under this finite probability model, the observed proportions determine the corresponding probabilities.

Table 1.1: Contingency Table for Data Hills Academy (frequencies)

	Pass	Fail	Total
Boy	450	150	600
Girl	350	50	400
Total	800	200	1000

Probability Types: A Complete Breakdown

1. Marginal Probabilities (Single Events)

These are the probabilities of individual events without conditioning on anything else.

$$P(\text{Boy}) = \frac{600}{1000} = 0.60 \quad (60\% \text{ are boys})$$

$$P(\text{Girl}) = \frac{400}{1000} = 0.40 \quad (40\% \text{ are girls})$$

$$P(\text{Pass}) = \frac{800}{1000} = 0.80 \quad (80\% \text{ passed})$$

$$P(\text{Fail}) = \frac{200}{1000} = 0.20 \quad (20\% \text{ failed})$$

2. Joint Probabilities (AND – Intersection)

These are the probabilities that **both events happen together**.

$$P(\text{Boy} \cap \text{Pass}) = \frac{450}{1000} = 0.45 \quad \text{Probability of a boy AND passed.}$$

$$P(\text{Boy} \cap \text{Fail}) = \frac{150}{1000} = 0.15 \quad \text{Probability of a boy AND failed.}$$

$$P(\text{Girl} \cap \text{Pass}) = \frac{350}{1000} = 0.35 \quad \text{Probability of a girl AND passed.}$$

$$P(\text{Girl} \cap \text{Fail}) = \frac{50}{1000} = 0.05 \quad \text{Probability of a girl AND failed.}$$

****Check:**** $0.45 + 0.15 + 0.35 + 0.05 = 1.00$, confirming all possibilities are covered.

3. Union Probabilities (OR)

Using the addition rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$:

$$P(\text{Boy} \cup \text{Pass}) = 0.60 + 0.80 - 0.45 = 0.95$$

$$P(\text{Girl} \cup \text{Pass}) = 0.40 + 0.80 - 0.35 = 0.85$$

$$P(\text{Boy} \cup \text{Fail}) = 0.60 + 0.20 - 0.15 = 0.65$$

$$P(\text{Girl} \cup \text{Fail}) = 0.40 + 0.20 - 0.05 = 0.55$$

4. Conditional Probabilities (Given)

These are probabilities calculated by restricting the sample space.

****Given gender $\rightarrow$ pass/fail:****

$$P(\text{Pass} \mid \text{Boy}) = \frac{0.45}{0.60} = 0.75 = 75\%$$

$$P(\text{Fail} \mid \text{Boy}) = \frac{0.15}{0.60} = 0.25 = 25\%$$

$$P(\text{Pass} \mid \text{Girl}) = \frac{0.35}{0.40} = 0.875 = 87.5\%$$

$$P(\text{Fail} \mid \text{Girl}) = \frac{0.05}{0.40} = 0.125 = 12.5\%$$

****Given pass/fail $\rightarrow$ gender:****

$$P(\text{Boy} \mid \text{Pass}) = \frac{0.45}{0.80} = 0.5625 = 56.25\%$$

$$P(\text{Girl} \mid \text{Pass}) = \frac{0.35}{0.80} = 0.4375 = 43.75\%$$

$$P(\text{Boy} \mid \text{Fail}) = \frac{0.15}{0.20} = 0.75 = 75\%$$

$$P(\text{Girl} \mid \text{Fail}) = \frac{0.05}{0.20} = 0.25 = 25\%$$

5. Bayes' Theorem (Posterior)

Bayes' theorem updates probabilities given new evidence. For example, the probability that a student is a girl given that they passed:

$$\begin{aligned} P(\text{Girl} \mid \text{Pass}) &= \frac{P(\text{Pass} \mid \text{Girl}) \cdot P(\text{Girl})}{P(\text{Pass})} \\ &= \frac{0.875 \times 0.40}{0.80} = \frac{0.35}{0.80} = 0.4375 = 43.75\% \end{aligned}$$

Other Bayes results:

$$P(\text{Boy} \mid \text{Pass}) = 56.25\% \quad P(\text{Girl} \mid \text{Pass}) = 43.75\%$$

$$P(\text{Boy} \mid \text{Fail}) = 75\% \quad P(\text{Girl} \mid \text{Fail}) = 25\%$$

Summary Table of All Probability Types

Table 1.2: Summary of Probability Types for Data Hills Academy

Type	Notation	Data Hills Example
Marginal	$P(A)$	$P(\text{Boy}) = 0.60$
Joint (AND)	$P(A \cap B)$	$P(\text{Boy} \cap \text{Pass}) = 0.45$
Union (OR)	$P(A \cup B)$	$P(\text{Boy} \cup \text{Pass}) = 0.95$
Conditional	$P(A B)$	$P(\text{Pass} \text{Girl}) = 0.875$
Bayes (Posterior)	$P(A B)$	$P(\text{Girl} \text{Pass}) = 0.4375$

Key Event Relationships

Using the Data Hills data, we can illustrate three fundamental event relationships:

- **Mutually Exclusive (Disjoint):** Events that cannot occur together. e.g., “Boy” and “Girl” have $P(\text{Boy} \cap \text{Girl}) = 0$.
- **Collectively Exhaustive:** Events whose union covers the entire sample space. e.g., $P(\text{Boy} \cup \text{Girl}) = 1$.
- **Independent:** Events where $P(A \cap B) = P(A)P(B)$. e.g., Check if “Girl” and “Pass” are independent: $P(\text{Girl} \cap \text{Pass}) = 0.35$, while $P(\text{Girl})P(\text{Pass}) = 0.40 \times 0.80 = 0.32$. Since $0.35 \neq 0.32$, they are **dependent**. Girls have a higher pass rate.

These concepts will be explored further in Chapter 4.

1.4 Real-World Application: Quality Control

In a factory producing 10,000 units daily, 150 are defective. If we select one unit at random, the probability it is defective is:

$$P(\text{Defective}) = \frac{150}{10000} = 0.015.$$

This empirical probability helps management decide whether to recalibrate machines.

Worked Example:

Quality Control Calculation: Total units = 10,000. Defective units = 150.

$$P(\text{Defective}) = \frac{\text{Number of defective units}}{\text{Total units}} = \frac{150}{10000} = 0.015.$$

As a percentage: $0.015 \times 100 = 1.5\%$. So the defect rate is 1.5%.

1.5 Exercises

Exercise (Basic Probability from Data Hills)

Using the Data Hills Academy data:

- (a) What is $\mathbb{P}(\text{Girl})$?
- (b) What is $\mathbb{P}(\text{Boy} \cap \text{Pass})$?
- (c) What is $\mathbb{P}(\text{Fail})$?

Solution (Full Workings):

- (a) $\mathbb{P}(\text{Girl}) = \frac{400}{1000} = 0.40.$
- (b) $\mathbb{P}(\text{Boy} \cap \text{Pass}) = \frac{450}{1000} = 0.45.$
- (c) $\mathbb{P}(\text{Fail}) = \frac{200}{1000} = 0.20.$

Chapter 2

Counting Techniques

2.1 Introduction

Before developing probability theory further, we need tools to count the number of outcomes in sample spaces and events. Counting techniques form the foundation of classical probability.

2.2 The Fundamental Counting Principle

If one event can occur in m ways and a second independent event can occur in n ways, then the two events together can occur in $m \times n$ ways.

Worked Example:

Simple Example: A restaurant offers 3 starters, 4 main courses, and 2 desserts. How many different full meals are possible?

$$3 \times 4 \times 2 = 24 \text{ meals.}$$

2.3 Factorials

The product of all positive integers up to n is called n factorial:

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 2 \times 1.$$

By definition, $0! = 1$.

Worked Example:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120.$$

2.4 Permutations

A permutation is an ordered arrangement of items. The number of ways to arrange r items from a set of n distinct items is:

$${}^n P_r = \frac{n!}{(n - r)!}.$$

Worked Example:

Example: How many ways can we arrange 3 books from a shelf of 5 different books?

$${}^5P_3 = \frac{5!}{(5-3)!} = \frac{120}{2!} = \frac{120}{2} = 60.$$

2.5 Combinations

A combination is an unordered selection of items. The number of ways to choose r items from a set of n distinct items is:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Worked Example:

Example: How many ways can we choose 3 students from a class of 10?

$$\binom{10}{3} = \frac{10!}{3!(10-3)!} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120.$$

2.6 Sampling with and without Replacement

- **With replacement:** Each selection is independent, and the same item can be chosen multiple times.
- **Without replacement:** Each item can be chosen at most once.

Worked Example:

Example: A bag contains 5 red and 3 blue balls. We draw 2 balls.

- **With replacement:** $8 \times 8 = 64$ possible ordered outcomes.
- **Without replacement:** $8 \times 7 = 56$ possible ordered outcomes.

2.7 Exercises

Exercise (Counting)

- How many ways can 4 students be arranged in a row?
- How many ways can we choose 2 cards from a standard 52-card deck?
- A PIN code has 4 digits. How many possible PINs are there if digits can repeat?

Solution (Full Workings):

(a) $4! = 4 \times 3 \times 2 \times 1 = 24$ ways.

(b) $\binom{52}{2} = \frac{52 \times 51}{2} = 1326$ ways.

(c) $10 \times 10 \times 10 \times 10 = 10,000$ possible PINs.

Chapter 3

Axiomatic Probability

3.1 Probability Spaces

In modern probability theory, we formalise the concept of a probability space.

Definition 3.1 (Probability Space). A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where:

- Ω is the sample space (set of all possible outcomes).
- $\mathcal{F}$ is a σ -algebra of events (a collection of subsets of Ω satisfying certain properties).
- $\mathbb{P}$ is a probability measure that assigns probabilities to events in $\mathcal{F}$.

3.2 Kolmogorov's Axioms

The modern theory of probability is built on three axioms, introduced by Andrey Kolmogorov in 1933.

Definition 3.2 (Kolmogorov Axioms). Let Ω be a sample space and let $\mathbb{P}$ be a probability measure defined on the events of Ω . Then:

A1. Non-negativity: For every event A ,

$$\mathbb{P}(A) \geq 0.$$

A2. Normalisation: The probability of the entire sample space is

$$\mathbb{P}(\Omega) = 1.$$

A3. Countable additivity: If $A_1, A_2, \dots$ are pairwise disjoint events, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Worked Example:

Verifying the Axioms with Data Hills:

1. Non-negativity: All probabilities from the data are between 0 and 1.
2. Normalisation: $\mathbb{P}(\text{Boy}) + \mathbb{P}(\text{Girl}) = 0.60 + 0.40 = 1.00$.
3. Additivity: The events "Boy" and "Girl" are disjoint, and their union is the whole sample space:

$$\mathbb{P}(\text{Boy} \cup \text{Girl}) = \mathbb{P}(\text{Boy}) + \mathbb{P}(\text{Girl}) = 0.60 + 0.40 = 1.00.$$

Insight 3.3. The three axioms are the starting point of probability theory. Many familiar rules—including the complement rule, addition rule, and several properties of conditional probability—can be derived from these basic principles.

Chapter 4

Relationships Between Events

4.1 Mutually Exclusive Events

Definition 4.1 (Mutually Exclusive). Two events A and B are mutually exclusive (disjoint) if they cannot occur simultaneously: $A \cap B = \emptyset$.

As noted in Chapter 1, “Boy” and “Girl” are mutually exclusive, so $\mathbb{P}(\text{Boy} \cap \text{Girl}) = 0$.

4.2 Independent Events

Definition 4.2 (Independent). Events A and B are independent if the occurrence of one does not affect the probability of the other:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

In Chapter 1 we tested “Girl” and “Pass” and found they are dependent because $\mathbb{P}(\text{Girl} \cap \text{Pass}) \neq \mathbb{P}(\text{Girl})\mathbb{P}(\text{Pass})$.

Insight 4.3. Independence is a stronger condition than mutual exclusivity. If two events are mutually exclusive and both have positive probability, they cannot be independent because knowing A occurred tells you B did not.

4.3 Pairwise vs. Mutual Independence

Definition 4.4 (Pairwise Independence). Events $A_1, A_2, \dots, A_n$ are pairwise independent if every pair satisfies:

$$\mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j) \quad \text{for all } i \neq j.$$

Definition 4.5 (Mutual Independence). Events $A_1, A_2, \dots, A_n$ are mutually independent if for every subset $\{i_1, \dots, i_k\}$,

$$\mathbb{P}\left(\bigcap_{j=1}^k A_{i_j}\right) = \prod_{j=1}^k \mathbb{P}(A_{i_j}).$$

Insight 4.6. Mutual independence is stronger than pairwise independence. Pairwise independence does not necessarily imply mutual independence.

4.4 Collectively Exhaustive Events

Definition 4.7 (Collectively Exhaustive). A collection of events $A_1, \dots, A_n$ is collectively exhaustive if at least one of them must occur, i.e., $\bigcup_{i=1}^n A_i = \Omega$.

In Data Hills, “Boy” and “Girl” are collectively exhaustive, as are “Pass” and “Fail”.

4.5 Complement of an Event

The complement of A , denoted A^c , is the event that A does not occur.

Worked Example:

Complement in Data Hills: If $A = \text{Pass}$, then $A^c = \text{Fail}$.

$$\mathbb{P}(\text{Pass}) = 0.80 \implies \mathbb{P}(\text{Fail}) = 1 - 0.80 = 0.20.$$

This matches the data: 200 out of 1000 students failed.

4.6 Real-World Application: Credit Risk

In banking, a loan can be “fully paid” or “defaulted” — these are mutually exclusive. The probability of default might be dependent on other factors like employment status.

4.7 Exercises

Exercise (Independence Test)

Using Data Hills, check whether “Girl” and “Pass” are independent.

Solution (Full Workings):

We need to verify if $\mathbb{P}(\text{Girl} \cap \text{Pass}) = \mathbb{P}(\text{Girl})\mathbb{P}(\text{Pass})$.

Step 1: From the data:

$$\mathbb{P}(\text{Girl} \cap \text{Pass}) = \frac{350}{1000} = 0.35.$$

Step 2: Compute individual probabilities:

$$\mathbb{P}(\text{Girl}) = \frac{400}{1000} = 0.40, \quad \mathbb{P}(\text{Pass}) = \frac{800}{1000} = 0.80.$$

Step 3: Multiply:

$$\mathbb{P}(\text{Girl}) \times \mathbb{P}(\text{Pass}) = 0.40 \times 0.80 = 0.32.$$

Step 4: Compare: $0.35 \neq 0.32$. Since they are not equal, the events are **dependent**. Girls have a higher pass rate than expected under independence.

Chapter 5

Fundamental Probability Rules

5.1 The Complement Rule

Theorem 5.1 (Complement Rule). *For any event A , $P(A^c) = 1 - P(A)$.*

Worked Example:

Data Hills Example: Let $A = \text{Pass}$. Then $A^c = \text{Fail}$.

$$P(\text{Pass}) = 0.80 \implies P(\text{Fail}) = 1 - 0.80 = 0.20.$$

This matches the data: 200 out of 1000 students failed.

5.2 The Addition Rule

Theorem 5.2 (General Addition Rule). *For any two events A and B ,*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are mutually exclusive, $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$.

Worked Example:

Data Hills Example: What is $P(\text{Boy} \cup \text{Pass})$?

Step 1: Identify the required probabilities:

$$P(\text{Boy}) = 0.60, \quad P(\text{Pass}) = 0.80, \quad P(\text{Boy} \cap \text{Pass}) = 0.45.$$

Step 2: Apply the addition rule:

$$\begin{aligned} P(\text{Boy} \cup \text{Pass}) &= P(\text{Boy}) + P(\text{Pass}) - P(\text{Boy} \cap \text{Pass}) \\ &= 0.60 + 0.80 - 0.45 \\ &= 0.95. \end{aligned}$$

Step 3: Interpret: There is a 95% chance that a randomly selected student is either a boy, passed, or both.

5.3 The Multiplication Rule

Theorem 5.3 (General Multiplication Rule). *For any events A and B ,*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B|A) = \mathbb{P}(B) \cdot \mathbb{P}(A|B).$$

If A and B are independent, $\mathbb{P}(B|A) = \mathbb{P}(B)$, so $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Worked Example:

Data Hills Example (Without Replacement): Two students are selected randomly without replacement. What is the probability that both are girls?

Step 1: Probability the first is a girl:

$$\mathbb{P}(\text{1st Girl}) = \frac{400}{1000} = 0.40.$$

Step 2: After removing one girl, there are 399 girls left out of 999 students:

$$\mathbb{P}(\text{2nd Girl} \mid \text{1st Girl}) = \frac{399}{999} \approx 0.3994.$$

Step 3: Apply the multiplication rule:

$$\mathbb{P}(\text{Both Girls}) = \mathbb{P}(\text{1st Girl}) \times \mathbb{P}(\text{2nd Girl} \mid \text{1st Girl}) = 0.40 \times 0.3994 \approx 0.1598.$$

Step 4: Interpret: There is approximately a 16% chance that both selected students are girls.

5.4 Real-World Application: System Reliability

An aircraft has two independent engines, each with reliability 0.99. The probability that at least one works is:

$$\mathbb{P}(\text{at least one}) = 1 - \mathbb{P}(\text{both fail}) = 1 - (0.01)^2 = 0.9999.$$

Worked Example:

System Reliability Calculation: Engine reliability = 0.99, so failure probability = $1 - 0.99 = 0.01$.

$$\mathbb{P}(\text{Engine 1 fails}) = 0.01, \quad \mathbb{P}(\text{Engine 2 fails}) = 0.01.$$

Since the engines are independent:

$$\mathbb{P}(\text{Both fail}) = 0.01 \times 0.01 = 0.0001.$$

Therefore:

$$\mathbb{P}(\text{At least one works}) = 1 - 0.0001 = 0.9999 = 99.99\%.$$

Redundancy dramatically reduces the probability of total system failure.

5.5 Exercises

Exercise (Addition Rule)

At Data Hills, what is $\mathbb{P}(\text{Boy} \cup \text{Pass})$?

Solution (Full Workings):

Using the general addition rule:

$$\begin{aligned}\mathbb{P}(\text{Boy} \cup \text{Pass}) &= \mathbb{P}(\text{Boy}) + \mathbb{P}(\text{Pass}) - \mathbb{P}(\text{Boy} \cap \text{Pass}) \\ &= 0.60 + 0.80 - 0.45 \\ &= 0.95.\end{aligned}$$

Thus $\boxed{\mathbb{P}(\text{Boy} \cup \text{Pass}) = 0.95}$.

Exercise (Multiplication Rule)

Two students are selected randomly without replacement. What is the probability that both are girls?

Solution (Full Workings):

$$\begin{aligned}\mathbb{P}(\text{1st Girl}) &= \frac{400}{1000} = 0.40. \\ \mathbb{P}(\text{2nd Girl} \mid \text{1st Girl}) &= \frac{399}{999} \approx 0.3994. \\ \mathbb{P}(\text{both girls}) &= 0.40 \times 0.3994 \approx 0.1598.\end{aligned}$$

Chapter 6

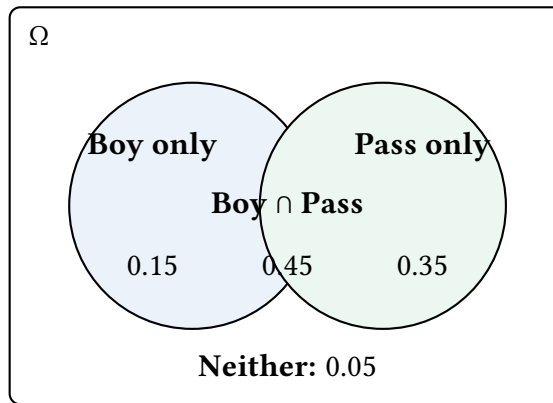
Visualising Probability

6.1 Venn Diagrams

Venn diagrams are a powerful tool for visualising relationships between events.

Worked Example:

Venn Diagram for Data Hills:



From the diagram:

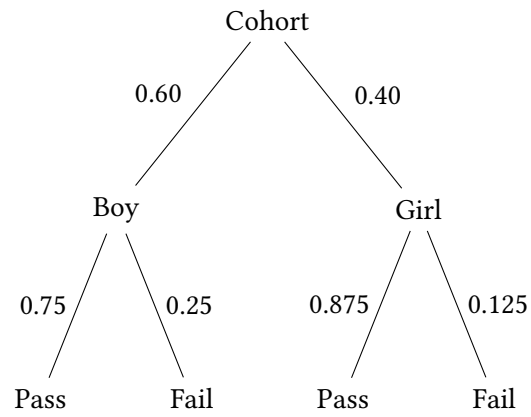
- Boy only: $0.60 - 0.45 = 0.15$ (boys who failed).
- Pass only: $0.80 - 0.45 = 0.35$ (girls who passed).
- Neither: $1.00 - (0.15 + 0.45 + 0.35) = 0.05$ (girls who failed).

6.2 Tree Diagrams

Tree diagrams decompose sequential processes.

Worked Example:

Tree Diagram for Data Hills:



From the tree:

- Probability of selecting a boy who passed: $0.60 \times 0.75 = 0.45$.
- Probability of selecting a girl who passed: $0.40 \times 0.875 = 0.35$.
- Total pass probability: $0.45 + 0.35 = 0.80$.

Chapter 7

Conditional Probability

7.1 Definition and Intuition

Conditional probability measures the chance of event A given that event B has occurred.

Definition 7.1 (Conditional Probability).

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad \text{provided } \mathbb{P}(B) > 0.$$

The calculations for Data Hills have already been shown in Chapter 1. We will now generalise the concept.

7.2 The Contingency Table (Probability Form)

Table 7.1: Data Hills Academy Contingency Table (Probabilities)

	Pass	Fail	Total
Boy	0.45	0.15	0.60
Girl	0.35	0.05	0.40
Total	0.80	0.20	1.00

From this table, we can read all joint, marginal, and conditional probabilities.

7.3 Real-World Application: Medical Testing

Sensitivity = $\mathbb{P}(T^+ | D^+)$, Specificity = $\mathbb{P}(T^- | D^-)$.

Worked Example:

Medical Test Example: A disease affects 1% of the population. A test has 95% sensitivity and 90% specificity.

Step 1: Identify known values:

$$\mathbb{P}(D) = 0.01, \quad \mathbb{P}(T^+ | D) = 0.95, \quad \mathbb{P}(T^- | D^-) = 0.90.$$

Step 2: Compute false positive rate:

$$\mathbb{P}(T^+ | D^-) = 1 - 0.90 = 0.10.$$

Step 3: The probability of a positive test is:

$$\mathbb{P}(T^+) = \mathbb{P}(T^+ | D)\mathbb{P}(D) + \mathbb{P}(T^+ | D^-)\mathbb{P}(D^-) = 0.95 \times 0.01 + 0.10 \times 0.99 = 0.0095 + 0.099 = 0.1085.$$

Step 4: Probability of disease given a positive test:

$$\mathbb{P}(D | T^+) = \frac{0.95 \times 0.01}{0.1085} = \frac{0.0095}{0.1085} \approx 0.0876.$$

Even with a positive test, there is only an 8.76% chance of having the disease because the disease is rare.

7.4 Exercises

Exercise (Conditional from Data Hills)

Find $\mathbb{P}(\text{Fail} | \text{Boy})$ and $\mathbb{P}(\text{Boy} | \text{Pass})$.

Solution (Full Workings):

For $\mathbb{P}(\text{Fail} | \text{Boy})$:

$$\mathbb{P}(\text{Fail} \cap \text{Boy}) = 0.15, \quad \mathbb{P}(\text{Boy}) = 0.60.$$

$$\mathbb{P}(\text{Fail} | \text{Boy}) = \frac{0.15}{0.60} = 0.25 = 25\%.$$

For $\mathbb{P}(\text{Boy} | \text{Pass})$:

$$\mathbb{P}(\text{Boy} \cap \text{Pass}) = 0.45, \quad \mathbb{P}(\text{Pass}) = 0.80.$$

$$\mathbb{P}(\text{Boy} | \text{Pass}) = \frac{0.45}{0.80} = 0.5625 = 56.25\%.$$

Chapter 8

The Law of Total Probability

8.1 Partitions

Before stating the law, we need the concept of a partition.

Definition 8.1 (Partition). A collection of events $A_1, \dots, A_k$ forms a partition of Ω if:

1. They are mutually exclusive: $A_i \cap A_j = \emptyset$ for $i \neq j$.
2. They are collectively exhaustive: $\bigcup_{i=1}^k A_i = \Omega$.

8.2 Statement and Derivation

If $A_1, \dots, A_k$ form a partition of Ω , then for any event B :

$$\mathbb{P}(B) = \sum_{i=1}^k \mathbb{P}(B \cap A_i) = \sum_{i=1}^k \mathbb{P}(B \mid A_i) \mathbb{P}(A_i).$$

Worked Example:

Data Hills Example: At Data Hills, suppose 70% are Day Scholars (85% pass) and 30% Boarders (65% pass). Find overall pass rate.

Step 1: Identify the partition:

$$\mathbb{P}(D) = 0.70, \quad \mathbb{P}(B) = 0.30.$$

Step 2: Identify conditional probabilities:

$$\mathbb{P}(\text{Pass} \mid D) = 0.85, \quad \mathbb{P}(\text{Pass} \mid B) = 0.65.$$

Step 3: Apply the law of total probability:

$$\mathbb{P}(\text{Pass}) = \mathbb{P}(\text{Pass} \mid D) \mathbb{P}(D) + \mathbb{P}(\text{Pass} \mid B) \mathbb{P}(B) = 0.85 \times 0.70 + 0.65 \times 0.30 = 0.595 + 0.195 = 0.79.$$

Step 4: Interpret: The overall pass rate is 79%.

8.3 Real-World Application: Spam Filtering

Three filters handle posts with different flag rates. The overall spam rate is computed by averaging across filters.

Worked Example:

Spam Filter Example: Filter A handles 50% of posts (flags 1% spam). Filter B handles 30% (flags 5% spam). Filter C handles 20% (flags 10% spam).

Step 1: Set up:

$$\begin{aligned} P(A) &= 0.50, & P(B) &= 0.30, & P(C) &= 0.20. \\ P(S | A) &= 0.01, & P(S | B) &= 0.05, & P(S | C) &= 0.10. \end{aligned}$$

Step 2: Apply law of total probability:

$$P(S) = 0.01 \times 0.50 + 0.05 \times 0.30 + 0.10 \times 0.20 = 0.005 + 0.015 + 0.020 = 0.04.$$

Step 3: Interpret: The overall spam flagging rate is 4%.

8.4 Exercises

Exercise (Total Probability)

At Data Hills, suppose 70% are Day Scholars (85% pass) and 30% Boarders (65% pass). Find overall pass rate.

Solution (Full Workings):

$$P(\text{Pass}) = P(\text{Pass} | D)P(D) + P(\text{Pass} | B)P(B) = 0.85 \times 0.70 + 0.65 \times 0.30 = 0.595 + 0.195 = 0.79.$$

The overall pass rate is 79%.

Chapter 9

Bayes' Theorem

9.1 The Theorem

Theorem 9.1 (Bayes' Theorem).

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(B | A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

When $A_1, \dots, A_k$ partition Ω ,

$$\mathbb{P}(A_j | B) = \frac{\mathbb{P}(B | A_j)\mathbb{P}(A_j)}{\sum_{i=1}^k \mathbb{P}(B | A_i)\mathbb{P}(A_i)}.$$

9.2 Terminology

- $\mathbb{P}(A)$: **Prior** — initial belief before observing evidence B .
- $\mathbb{P}(B | A)$: **Likelihood** — probability of observing evidence B if hypothesis A is true.
- $\mathbb{P}(B)$: **Evidence** — total probability of observing evidence B .
- $\mathbb{P}(A | B)$: **Posterior** — updated probability of hypothesis A given evidence B .

The Data Hills Bayes examples were already worked in Chapter 1. We now present a new medical application.

9.3 Real-World Application: Medical Diagnosis

Given a positive test result, what is the probability the patient actually has the disease?

Worked Example:

Classic Medical Example: Disease prevalence 1%, sensitivity 95%, false positive rate 5%.

Step 1: Identify priors and likelihoods:

$$\begin{aligned} \mathbb{P}(D) &= 0.01, & \mathbb{P}(D') &= 0.99, \\ \mathbb{P}(T^+ | D) &= 0.95, & \mathbb{P}(T^+ | D') &= 0.05. \end{aligned}$$

Step 2: Compute evidence using law of total probability:

$$\mathbb{P}(T^+) = \mathbb{P}(T^+ | D)\mathbb{P}(D) + \mathbb{P}(T^+ | D')\mathbb{P}(D') = 0.95 \times 0.01 + 0.05 \times 0.99 = 0.0095 + 0.0495 = 0.059.$$

Step 3: Apply Bayes' theorem:

$$\mathbb{P}(D \mid T^+) = \frac{\mathbb{P}(T^+ \mid D)\mathbb{P}(D)}{\mathbb{P}(T^+)} = \frac{0.95 \times 0.01}{0.059} = \frac{0.0095}{0.059} \approx 0.161.$$

Step 4: Interpret: Even with a positive test, there is only a 16.1% chance of having the disease because the disease is rare.

This example illustrates the importance of the base rate and the common **base-rate fallacy**, in which the prevalence of a condition is ignored when interpreting diagnostic evidence.

9.4 Exercises

Exercise (Bayes – Classic Medical)

Disease prevalence 1%, sensitivity 95%, false positive rate 5%. Compute $\mathbb{P}(D \mid T^+)$.

Solution (Full Workings):

$$\mathbb{P}(T^+) = 0.95 \times 0.01 + 0.05 \times 0.99 = 0.0095 + 0.0495 = 0.059.$$

$$\mathbb{P}(D \mid T^+) = \frac{0.95 \times 0.01}{0.059} = \frac{0.0095}{0.059} \approx 0.161.$$

There is a 16.1% chance of having the disease given a positive test.

Part II

RANDOM VARIABLES AND DISTRIBUTIONS

Chapter 10

Random Variables

10.1 Definition and Motivation

Often, we are not interested in the raw outcomes of a random experiment, but rather in some numerical function of those outcomes.

Definition 10.1 (Random Variable). A random variable X is a function that assigns a real number to each outcome in the sample space:

$$X : \Omega \rightarrow \mathbb{R}.$$

Worked Example:

Coin Toss Example: Let $\Omega = \{H, T\}$. Define X as the number of heads:

$$X(H) = 1, \quad X(T) = 0.$$

This is a random variable.

10.2 Discrete Random Variables

A random variable is discrete if it takes a finite or countably infinite set of values.

Definition 10.2 (Probability Mass Function (PMF)). For a discrete random variable X , the PMF is:

$$p_X(x) = \mathbb{P}(X = x).$$

Properties:

1. $p_X(x) \geq 0$ for all x .
2. $\sum_x p_X(x) = 1$.

Worked Example:

Data Hills Example: Define X as the number of students selected who are girls when we select 2 students without replacement. X can take values 0, 1, 2. The PMF is:

$$p_X(0) = \frac{600}{1000} \times \frac{599}{999} \approx 0.3598,$$

$$p_X(2) = \frac{400}{1000} \times \frac{399}{999} \approx 0.1598,$$

$$p_X(1) = 1 - p_X(0) - p_X(2) \approx 0.4804.$$

10.3 Continuous Random Variables

A random variable is continuous if it can take any value in an interval.

Definition 10.3 (Probability Density Function (PDF)). For a continuous random variable X , the PDF $f_X(x)$ satisfies:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx,$$

and

$$\int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Worked Example:

Uniform Example: If $X \sim \text{Uniform}(0, 1)$, then:

$$f_X(x) = 1 \quad \text{for } 0 \leq x \leq 1.$$

Then $\mathbb{P}(0.2 \leq X \leq 0.5) = \int_{0.2}^{0.5} 1 dx = 0.3$.

10.4 Exercises

Exercise (Random Variable)

A fair die is rolled. Let X be the number showing.

- (a) What is the PMF of X ?
- (b) What is $\mathbb{P}(X \geq 4)$?

Solution (Full Workings):

(a) $p_X(x) = \frac{1}{6}$ for $x = 1, 2, 3, 4, 5, 6$. (b) $\mathbb{P}(X \geq 4) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$.

Chapter 11

Discrete Probability Distributions

11.1 The Bernoulli Distribution

A Bernoulli random variable takes value 1 with probability p and 0 with probability $1 - p$.

Definition 11.1 (Bernoulli Distribution). $X \sim \text{Bernoulli}(p)$ has PMF:

$$p_X(1) = p, \quad p_X(0) = 1 - p.$$

11.2 The Binomial Distribution

A binomial random variable counts the number of successes in n independent Bernoulli trials.

Definition 11.2 (Binomial Distribution). $X \sim \text{Binomial}(n, p)$ has PMF:

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Worked Example:

Example: A student guesses on a 10-question multiple-choice test (each with 4 options). What is the probability of getting exactly 3 correct?

$$\mathbb{P}(X = 3) = \binom{10}{3} (0.25)^3 (0.75)^7 \approx 0.2503.$$

11.3 The Poisson Distribution

A Poisson random variable counts the number of events in a fixed interval.

Definition 11.3 (Poisson Distribution). $X \sim \text{Poisson}(\lambda)$ has PMF:

$$\mathbb{P}(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Worked Example:

Example: A call centre receives $\lambda = 5$ calls per hour. What is the probability of receiving exactly 3 calls in an hour?

$$P(X = 3) = \frac{e^{-5}5^3}{3!} = \frac{0.006738 \times 125}{6} \approx 0.1404.$$

11.4 Exercises

Exercise (Binomial Distribution)

A fair coin is tossed 8 times. Let X be the number of heads.

- (a) Find $P(X = 3)$.
- (b) Find $P(X \geq 5)$.

Solution (Full Workings):

(a) $\binom{8}{3}(0.5)^3(0.5)^5 = 56 \times (0.5)^8 = 56/256 \approx 0.21875$. (b) $P(X \geq 5) = P(X = 5) + P(X = 6) + P(X = 7) + P(X = 8)$. $\binom{8}{5} = 56$, $\binom{8}{6} = 28$, $\binom{8}{7} = 8$, $\binom{8}{8} = 1$. $(56 + 28 + 8 + 1)/256 = 93/256 \approx 0.3633$.

Chapter 12

Continuous Probability Distributions

12.1 The Uniform Distribution

Definition 12.1 (Uniform Distribution). $X \sim \text{Uniform}(a, b)$ has PDF:

$$f_X(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

12.2 The Exponential Distribution

Definition 12.2 (Exponential Distribution). $X \sim \text{Exponential}(\lambda)$ has PDF:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

12.3 The Normal Distribution

The normal distribution is the most important continuous distribution.

Definition 12.3 (Normal Distribution). $X \sim N(\mu, \sigma^2)$ has PDF:

$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Worked Example:

Standard Normal: If $Z \sim N(0, 1)$, we can compute probabilities using the standard normal table.

$$\mathbb{P}(Z \leq 1.96) \approx 0.975.$$

Chapter 13

Expectation and Variance

13.1 Expected Value

The expected value is the long-run average of a random variable.

Definition 13.1 (Expected Value). For a discrete random variable X with PMF $p_X(x)$:

$$\mathbb{E}[X] = \sum_x x p_X(x).$$

For a continuous random variable X with PDF $f_X(x)$:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

Worked Example:

Binomial Example: If $X \sim \text{Binomial}(10, 0.5)$, then $\mathbb{E}[X] = 10 \times 0.5 = 5$.

13.2 Variance

The variance measures the spread of a random variable around its mean.

Definition 13.2 (Variance).

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Worked Example:

Binomial Example: If $X \sim \text{Binomial}(10, 0.5)$, then $\text{Var}(X) = 10 \times 0.5 \times 0.5 = 2.5$.

Chapter 14

Limit Theorems

14.1 Law of Large Numbers (LLN)

The LLN states that the sample average converges to the expected value as the sample size grows.

Theorem 14.1 (Law of Large Numbers). *If $X_1, X_2, \dots$ are iid with $E[X_i] = \mu$, then:*

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{prob}} \mu.$$

14.2 Central Limit Theorem (CLT)

The CLT is the most important result in probability for statistics.

Theorem 14.2 (Central Limit Theorem). *If $X_1, X_2, \dots, X_n$ are iid with mean μ and variance σ^2 , then for large n :*

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{\text{dist}} N(0, 1).$$

Worked Example:

Example: A population has $\mu = 100$, $\sigma = 15$. For $n = 36$:

$$P(\bar{X} \leq 105) \approx P\left(Z \leq \frac{105 - 100}{15/\sqrt{36}}\right) = P(Z \leq 2) \approx 0.9772.$$

Appendix A

Formula Summary

Core Formulas

Concept	Formula
Probability	$P(A) = \frac{ A }{ \Omega }$
Complement	$P(A^c) = 1 - P(A)$
Union	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional	$P(A B) = \frac{P(A \cap B)}{P(B)}$
Multiplication	$P(A \cap B) = P(A)P(B A)$
Independence	$P(A \cap B) = P(A)P(B)$
Total Probability	$P(B) = \sum_i P(B A_i)P(A_i)$
Bayes	$P(A B) = \frac{P(B A)P(A)}{P(B)}$
Expected Value	$E[X] = \sum_x x p_X(x)$
Variance	$\text{Var}(X) = E[X^2] - (E[X])^2$
Binomial PMF	$\binom{n}{k} p^k (1-p)^{n-k}$
Poisson PMF	$\frac{e^{-\lambda} \lambda^k}{k!}$
Normal PDF	$\frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$

Appendix B

Glossary of Terms

Central Limit Theorem The distribution of sample means approaches normality.

Conditional Probability Probability of an event given another has occurred.

Event A subset of the sample space.

Expected Value The long-run average of a random variable.

Independent Occurrence of one event does not affect another.

Mutually Exclusive Two events that cannot occur together.

Probability Space $(\Omega, \mathcal{F}, P)$.

Random Variable A function from the sample space to $\mathbb{R}$.

Sample Space The set of all possible outcomes.

Variance Measure of spread around the mean.

Appendix C

Solutions to All Exercises

All solutions are provided in the preceding chapters. Refer to the relevant sections.